

LITERATURE REVIEW

Out-of-Distribution Detection for Non-Image Numerical Time-Series Data in Social Infrastructure AI Systems

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Abstract

This review concerns only non-image numerical time-series data used in social-infrastructure monitoring and control. Typical inputs are sequences of temperatures, pressures, flows, voltages, valve positions, vibration values, server KPIs, or spacecraft telemetry. Image, video, text, and audio OOD benchmarks are outside the evaluation scope.

For these systems, OOD detection means determining whether a numerical time window is insufficiently represented by the normal operating data or by the classes used to train a temporal model. The appropriate technical family depends on the available model and labels. For normal-only process monitoring, statistical process monitoring, change detection, one-class learning, forecasting residuals, and reconstruction models are the main choices. If a numerical time-series classifier already exists, distance or density modeling of its temporal features can be added. Recent time-series evidence suggests that deep-feature modeling is more reliable than directly transferring confidence heuristics developed for image classifiers [1].

The recommended benchmark stack is also entirely numerical: Tennessee Eastman Process (TEP), NASA SMAP/MSL telemetry, Server Machine Dataset (SMD), SWaT/WADI process-control data, and selected numerical datasets from the UEA multivariate archive. No image dataset is proposed. Candidate initial methods are PCA with Hotelling's T2 and SPE/Q statistics, Mahalanobis distance, Isolation Forest or One-Class SVM on window features, a forecasting/reconstruction baseline, and temporal deep-feature modeling. The concrete experimental scope should be determined by the tasks and data available during the internship.

1. Scope and Problem Definition

Let a numerical input window be $X_t = [x_{t-L+1}, \dots, x_t] \in \mathbb{R}^{L \times d}$, where L is the window length and d is the number of sensor or control channels. A detector produces a score $s(X_t)$, and an input is rejected or alerted when $s(X_t) > \tau$. The threshold τ must be calibrated on validation data that are separate from the final test data.

This review includes:

- multivariate sensor and actuator streams from industrial or infrastructure systems;
- telemetry and performance indicators from spacecraft, servers, networks, power, water, transport, and manufacturing systems;
- numerical time-series classification models for operating mode, event, or fault type;
- normal-only monitoring when abnormal labels are scarce.

This review excludes image, video, natural-language, and audio OOD detection. Methods originating in image research are not treated as primary candidates unless their mechanism can be applied directly to a numerical temporal model; even then, they are secondary comparison baselines rather than the center of the report.

2. Three Numerical Time-Series Settings

2.1 Semantic OOD for a time-series classifier

The model is trained on known numerical sequence classes, such as known operating modes or known fault categories. At test time, a window may belong to an unseen class. A detector uses temporal embeddings, logits, or class-conditional distributions to decide whether to reject the classifier output. A realistic benchmark holds out classes from the same original numerical dataset so that ID and OOD windows share sensors, units, sampling characteristics, and sequence length [1].

2.2 Operating-condition shift and concept drift

The system remains physically valid, but load, season, controller settings, calibration, equipment age, or environmental conditions change. This is often a gradual or persistent distribution shift rather than a short abnormal event. Change-point detection, drift statistics, adaptive reference windows, and regime-specific calibration are particularly relevant.

2.3 Time-series anomaly detection

The goal is to detect abnormal points or intervals relative to normal temporal dynamics. This setting is common when only normal data are available. It overlaps operationally with OOD detection but is usually evaluated using event labels, detection delay, and false alarms over time rather than only sample-level class rejection.

3. Characteristics of Infrastructure Time Series

- Cross-channel correlation: pressure, flow, valve position, and controller output may be individually plausible but jointly inconsistent.
- Temporal dependence: abnormality can occur in lag, derivative, phase, frequency, or transition structure rather than absolute value.
- Multiple normal regimes: startup, shutdown, maintenance, peak load, seasonal operation, and degraded-but-safe operation should not automatically be treated as OOD.
- Non-stationarity: the normal reference distribution changes over time, so static thresholds can produce alert floods.
- Irregular and missing data: sensor dropout, asynchronous sampling, and resampling can themselves change the score distribution.
- Contaminated training data: presumed-normal archives may contain small faults, manual interventions, or unrecorded maintenance.
- Operational cost: false alarms, late detection, edge-device compute limits, and explainability can matter more than a small AUROC improvement.

4. Methods for Non-Image Numerical Time Series

Method family	Representative techniques	Numerical time-series score	Strengths	Limitations	Suitable systems
Statistical process monitoring	PCA/DPCA, PLS, Hotelling's T2, SPE/Q [2-4]	Distance inside the retained latent subspace and residual energy outside it	Mature, fast, interpretable, strong industrial baseline	Linear/stationary assumptions; regime handling needed	Process plants, utilities, multichannel control data
Distance and robust covariance	Mahalanobis, robust covariance, class-conditional Gaussian	Distance from ID window features or temporal embeddings	Transparent and low-latency	Covariance estimation is difficult in high dimension or multimodal ID	Existing feature pipeline or temporal classifier
One-class and isolation methods	One-Class SVM, Isolation Forest, LOF, Deep SVDD [5]	Boundary, isolation depth, local density, or hypersphere distance	Normal-only training; works with engineered window features	Sensitive to scaling and feature design; local methods scale poorly	Equipment condition monitoring and moderate-dimensional windows
Sequential change detection	CUSUM, EWMA, Page-Hinkley, Bayesian/change-point methods	Accumulated deviation from a reference process	Detects persistent small shifts and supports online operation	Requires reference stability and careful reset/drift logic	Slow degradation, calibration drift, operating-condition changes
Forecasting residual	AR/ARIMA, state-space/Kalman, Gaussian process, LSTM/TCN/Transformer forecaster	Prediction residual, standardized innovation, or predictive likelihood	Directly models temporal dynamics; residuals can identify channels	Forecast error rises under legitimate regime changes; multi-step calibration is hard	Regularly sampled telemetry and control loops
Reconstruction models	PCA reconstruction, autoencoder, VAE, USAD [6]	Reconstruction error or reconstruction probability	Natural for normal-only multivariate data; variable-level scores possible	Over-generalization may reconstruct anomalies; contaminated training harms boundaries	IT monitoring, industrial sensors, water systems
Probabilistic temporal models	GMM, HMM, stochastic RNN/OmniAnomaly [7], normalizing flow	Negative log-likelihood or likelihood ratio	Represents uncertainty and multimodal dynamics	Raw likelihood can be misleading; distribution assumptions and calibration matter	Stochastic telemetry with repeated regimes
Temporal association/attention	Anomaly Transformer [8], TranAD [9], MEMTO [10]	Association discrepancy, self-conditioned reconstruction, latent/input deviation	Captures long-range and cross-variable dependence	More compute and tuning; published rankings are sensitive to evaluation protocol	High-dimensional multivariate telemetry with sufficient training data
Time-series representation + feature model	TS2Vec [11], temporal encoder + PCA/OCSVM/IF, DFM [1]	Distance or reconstruction error in a learned temporal	Separates representation learning from detector; promising	Depends on windowing, augmentation, and embedding	Semantic OOD for numerical sequence classifiers

Method family	Representative techniques	Numerical time-series score	Strengths	Limitations	Suitable systems
		embedding	in TS-OOD	stability	
Time-series-specific calibrated OOD	Seasonal Ratio Scoring [12], CODiT [13]	Seasonal likelihood ratio or conformal temporal-equivariance score	Uses temporal structure; conformal scores can support false-alarm interpretation	Seasonal decomposition may be costly; temporal dependence challenges calibration guarantees	Strongly periodic systems or safety cases requiring calibrated alarms

4.1 Statistical baselines should be mandatory

PCA-based process monitoring partitions a multivariate measurement into a principal-component subspace and a residual subspace. Hotelling's T2 monitors variation represented inside the model, while SPE/Q monitors lack of fit outside the model [2-4]. Dynamic PCA adds lagged variables or moving windows to capture serial dependence. These methods are computationally small, auditable, and well aligned with industrial numerical data; every deep method should be compared against them.

4.2 Window-feature one-class detection

A practical model-independent approach converts each numerical window into features such as mean, standard deviation, slope, RMS, derivative energy, spectral power, autocorrelation, and cross-channel correlation. Mahalanobis distance, One-Class SVM, or Isolation Forest is then fitted using ID windows only. This approach is suitable when the production AI model is inaccessible or when a transparent edge implementation is required.

4.3 Forecasting and reconstruction

Forecasting methods identify windows whose future values cannot be predicted from recent history. State-space innovations are particularly interpretable for control systems. Reconstruction methods learn normal-window structure and flag large residuals. USAD uses adversarially trained autoencoders for multivariate time series [6]; OmniAnomaly uses a stochastic recurrent model and reconstruction probability [7]. Their main risk is over-generalization: a powerful model may also predict or reconstruct an abnormal pattern.

4.4 Temporal deep models and learned features

Anomaly Transformer uses differences in temporal association structure [8]. TranAD uses transformer encoders, self-conditioning, and adversarial training for multivariate anomaly detection [9]. MEMTO adds a memory mechanism and scores deviation in input and latent spaces [10]. TS2Vec learns multiscale timestamp and subsequence representations that can feed a lightweight one-class detector [11].

For semantic OOD, TS-OOD compared numerical multivariate sequence datasets and found that many generic post-hoc confidence methods were unreliable, while deep-feature modeling with PCA or one-

class models was more promising [1]. Accordingly, learned temporal embedding plus a transparent feature detector is a higher-priority candidate than image-oriented activation clipping or weight sparsification.

5. Numerical Public Datasets

Priority	Dataset	Numerical channels and domain	Recommended role	Access / caveat
A	Tennessee Eastman Process (TEP) [14]	Simulated chemical-process measurements and manipulated variables with known faults	Industrial fault/OOD, operating-mode, and channel-attribution experiments	Public; synthetic and extensively studied, so real-world generalization must not be assumed
A	SMAP/MSL [15]	NASA spacecraft telemetry with labeled anomaly intervals	Telemetry anomaly detection, delay, and cross-channel evaluation	Public; preprocessing versions and interval labels must be standardized
A	Server Machine Dataset (SMD) [7]	38-dimensional KPIs from 28 server machines	Infrastructure monitoring and cross-machine robustness	Public; machines should be modeled separately unless transfer is explicit
A	UEA numerical multivariate subsets [1,16]	Sensor, motion, EEG/ECG, traffic, and other numerical sequences	Same-source class-held-out semantic OOD	Public; select numerical sensor datasets and exclude any image/video-derived task
B	SWaT / WADI [17]	Water-treatment and water-distribution sensors, actuators, and attacks	Cyber-physical and process-control validation	Request-based; redistribution is prohibited and attacks are not the complete OOD space
B	TSB-AD [18]	Large curated collection of numerical time-series anomaly datasets and algorithms	Metric stress test and broad anomaly comparison	Public; mostly TSAD rather than semantic class rejection
B	MTAD benchmark [19]	Public multivariate KPI datasets with unified methods and practical metrics	Accuracy, efficiency, delay, and salience comparison	Public toolkit; dataset preprocessing must be version-controlled
C	UCR Time Series Anomaly Archive [20]	Curated univariate numerical series	Sanity checks for metrics and thresholding	Public; limited multivariate control-system realism

There is no single universal numerical dataset. A sensible stack is: UEA numerical class-held-out data for semantic OOD; TEP for controlled industrial faults; SMAP/MSL or SMD for real telemetry; and SWaT/WADI only when access and usage conditions are acceptable. No image dataset is required or recommended.

6. Evaluation Protocol for Numerical Time Series

6.1 Splitting and calibration

1. Split by class, machine, operating regime, physical episode, or time block before creating overlapping windows.
2. Keep adjacent windows from one event in one split to avoid temporal leakage.
3. Fit normal-only detectors on ID training data. If outlier exposure is used, report it as a separate supervised condition.
4. Set thresholds using an independent ID calibration period or a clearly defined validation protocol; do not tune on final OOD labels.
5. Test near-OOD, far-OOD, gradual drift, abrupt sensor faults, and legitimate regime changes separately.
6. Report results across multiple machines, class-held-out splits, time periods, and random seeds.

6.2 Metrics

- AUROC and AUPR-OOD for ranking performance;
- FPR95 and false alarms per hour/day for operational alert cost;
- event recall, time-to-detect, and detection delay for interval anomalies;
- ID task accuracy or forecast error to ensure that the base model remains useful;
- inference latency, memory, and retraining time for edge or real-time deployment;
- threshold stability across seasons, loads, machines, and maintenance periods;
- variable-level attribution accuracy when fault localization is required.

Point-adjusted F1 must not be the only result. Research has shown that point adjustment can greatly inflate time-series anomaly scores and even distort method rankings [21]. Use unadjusted point metrics, event-based metrics, and delay-aware or range-aware metrics such as VUS-PR where appropriate [18,21].

7. Candidate Follow-up Evaluation Options

The following are candidate options, not a fixed experiment plan. The exact scope, datasets, and methods should be selected according to internship assignments, data access, and implementation time.

Candidate path	Methods	Suitable data	Purpose
Transparent process baseline	PCA/DPCA with T2 and SPE/Q; robust Mahalanobis	TEP, SWaT/WADI	Establish an interpretable industrial baseline
Model-independent window baseline	Window features + Isolation Forest or One-Class SVM	SMD, SMAP/MSL, TEP	Evaluate normal-only OOD without modifying an AI model

Candidate path	Methods	Suitable data	Purpose
Temporal prediction baseline	State-space/Kalman or compact TCN/LSTM forecaster	Regular telemetry or control data	Detect dynamic inconsistency and estimate channel residuals
Reconstruction baseline	PCA reconstruction or compact autoencoder/USAD	Multivariate sensor windows	Compare linear and nonlinear normal-pattern learning
Semantic numerical OOD	1-D temporal encoder + Mahalanobis or DFM-PCA/OCSVM	Selected UEA numerical datasets	Detect unseen sequence classes from the same sensors
Calibrated alarm option	Quantile or conformal thresholding with blocked calibration	One selected infrastructure dataset	Relate threshold choice to false-alarm behavior

As a possible follow-up, a transparent numerical pipeline could generate multichannel sensor windows, extract temporal and cross-channel features, fit an ID-only PCA subspace, and calibrate the threshold using held-out ID data.

8. Deployment Guidance

Process plants and utilities

Begin with regime-aware PCA/DPCA, T2, SPE/Q, and contribution analysis. Treat startup, shutdown, maintenance, and known load zones as separate valid regimes where necessary. Add a nonlinear or deep model only after the statistical baseline and threshold behavior are understood.

Telemetry and IT infrastructure

Use per-machine or per-entity models unless cross-entity transfer is an explicit goal. Forecasting residuals, compact autoencoders, or window-feature one-class models are practical first choices. Monitor missing-data rate and sampling changes as separate data-quality signals.

Numerical time-series classifiers

When a model predicts operating mode or fault type from numerical sequences, extract its pre-output temporal embedding and fit Mahalanobis, PCA reconstruction, One-Class SVM, or Isolation Forest on known-class embeddings. Energy or softmax scores may be included only as secondary baselines.

Safety and control integration

OOD detection should trigger abstention, operator review, or a predefined safe fallback. It should not directly change control actions without process constraints and safety validation. Use hysteresis and minimum-duration logic carefully, and report the detection delay they introduce.

9. Conclusions

This review focuses exclusively on non-image numerical time-series data. The most relevant methods are statistical process monitoring, one-class and isolation models, sequential change detection, forecasting residuals, reconstruction/probabilistic temporal models, and temporal representation plus feature-space detection. Image-oriented OOD benchmarks and image-specific activation methods are not part of the proposed evaluation.

As follow-up candidates, the team may consider PCA/DPCA with T2 and SPE/Q, robust Mahalanobis, window-feature Isolation Forest or One-Class SVM, a compact forecasting or reconstruction model, and deep-feature modeling for an existing numerical sequence classifier. These are not fixed experiments. The concrete experimental range should be determined according to the internship tasks, available numerical data, required system latency, and time available for implementation.

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